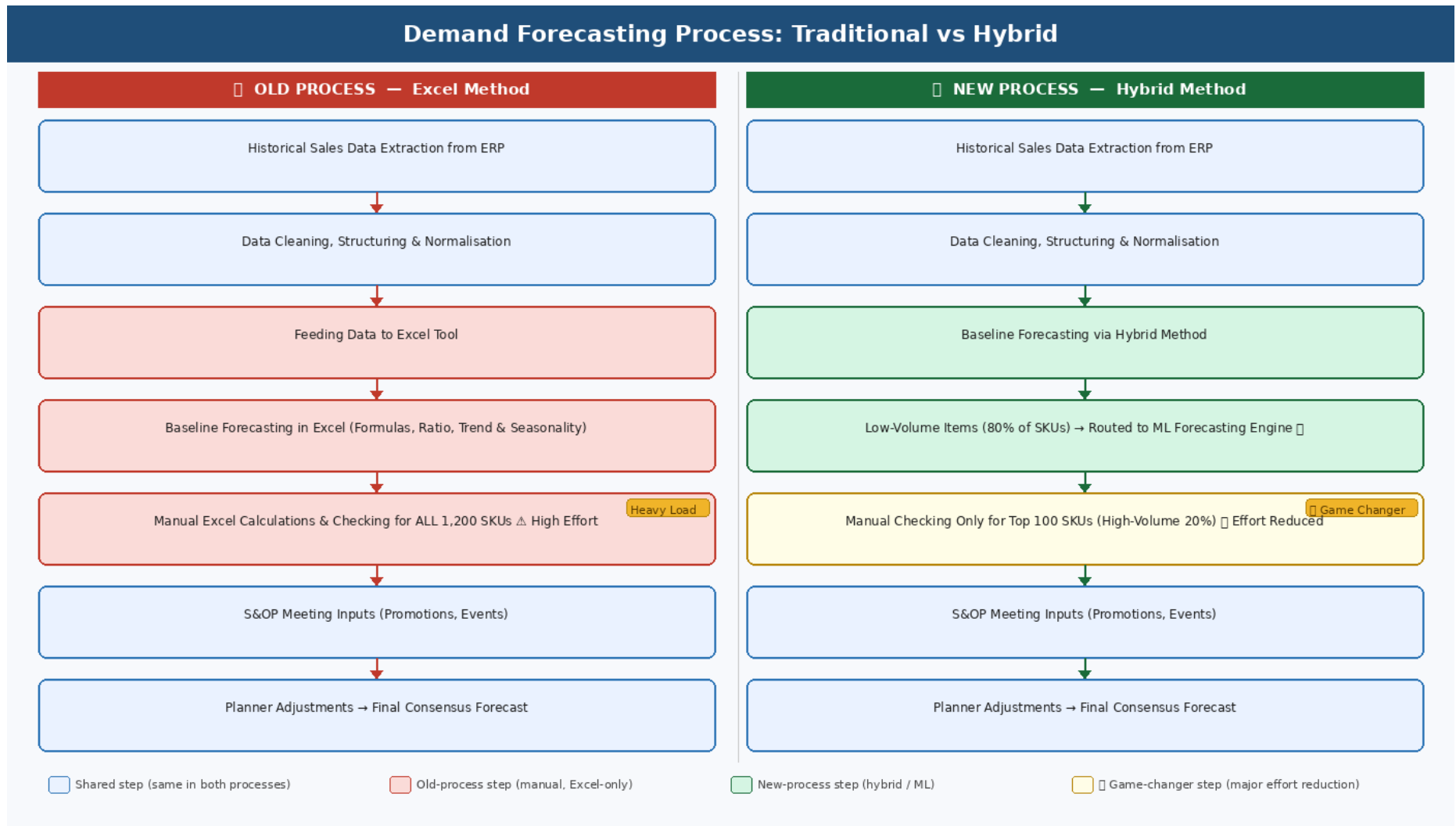


Executive summary

As part of a process improvement project conducted at AMP, a hybrid demand forecasting tool was designed and implemented to address the limitations of a fully manual, Excel-based forecasting process that required significant time and effort. The below workflow diagram shows how I calculated demand forecasting with the traditional method vs the new improved method.



ML-Enhanced Demand Forecasting Project

Hybrid Excel + XGBoost Forecasting Workflow for Supply Chain Planning

Organization	Primary Model	Business Output	Measured Impact
AMP	XGBoost	Power BI + Excel	Improved +4% forecast accuracy

1. Executive Summary

This case study presents a demand forecasting tool designed for AMP. The project combined machine learning, Excel-based planning logic and Power BI reporting to improve SKU-level forecast accuracy while keeping planner expertise in the process. The original planning workflow was highly manual, spreadsheet-heavy and difficult to scale across a large product portfolio. The hybrid solution used XGBoost to automate baseline forecasting, while planners focused on high-impact exceptions identified through Pareto segmentation.

2. Problem and Objective

Problem: Manual Excel forecasting required planners to review many SKU-level forecasts, creating scalability, consistency and decision-focus challenges.

Objective: Design a hybrid process that uses machine learning for repeatable forecasting and planner judgment for high-impact business exceptions.

- Reduce low-value manual work across 1,000+ SKUs.
- Improve forecast reliability without removing S&OP context.
- Create a scalable SKU-level workflow for planners and SCM teams.

3. Collaboration Model

My role	IT team	SCM team
Research, model selection, scientific methodology, hybrid design, Excel planning layer and implementation support.	ML pipeline, data ingestion, model execution environment and Power BI dashboard build.	Planning validation, S&OP inputs, exception review and business-rule feedback.

Confidentiality: source code, production data, real SKU/customer identifiers and proprietary logic are excluded.

4. Methodology and Data Flow

The methodology combined quantitative model evaluation with qualitative business validation. I compared classical forecasting approaches and machine learning options, then proposed a practical hybrid workflow that balances automation with planner judgment.

Step	Purpose	Assumed Techniques	Deliverable
Data understanding	Review demand history and planning pain points	SKU segmentation, historical demand review, stakeholder inputs	Planning requirements
Model selection	Select model for structured nonlinear data	XGBoost, rolling backtest, baseline comparison	Selected ML approach
Hybrid design	Combine automation and business review	ML forecast, Excel baseline, forecast combination, Pareto exceptions	Operational workflow
Evaluation	Measure accuracy and usability	MAPE, MAE, RMSE, actual vs forecast, planner feedback	Impact view

Sample Input Data Format (Anonymized)

AMPXX | Forecasting Data Template

Data Source: SFTP Excel / DWH Extract | Granularity: SKU x Market x Month | Horizon: 12 months

Period	Market	SKU_ID	Category	Lifecycle	Hist_Demand	Stock_Flag	Seasonality	Funnel_Index	Promo_Flag	Target_Qty
2025-01	FR-East	SKU-1000	Legacy	Phase-Out	582	OK	0.88	1.21	1	1069
2025-02	FR-East	SKU-1001	Legacy	Active	719	Backlog	1.29	1.23	1	386
2025-03	FR-East	SKU-1002	Premium	Phase-In	379	OK	0.81	0.80	0	1625
2025-04	FR-East	SKU-1003	Rechargeable	Phase-Out	454	Backlog	0.90	1.29	0	1622
2025-05	DE-West	SKU-1004	Accessory	Stable	610	Limited	1.22	0.85	0	1840
2025-06	FR-East	SKU-1005	Rechargeable	Stable	1767	Limited	0.86	0.88	0	1262
2025-07	IT-North	SKU-1006	Rechargeable	Replacement	439	OK	0.96	1.37	0	1145
2025-08	IT-North	SKU-1007	Legacy	Active	1208	Limited	1.07	0.85	0	855
2025-09	IT-South	SKU-1008	Rechargeable	Replacement	332	OK	1.29	1.24	0	150
2025-10	DE-West	SKU-1009	Legacy	Replacement	1779	OK	1.18	1.25	0	288

Features engineered from historical demand, product attributes, lifecycle status, availability, seasonality and funnel signals.

All values are synthetic. Column names are representative of the assumed input structure and do not expose confidential AMPXX data.

Figure 1. Representative input data template with synthetic values.

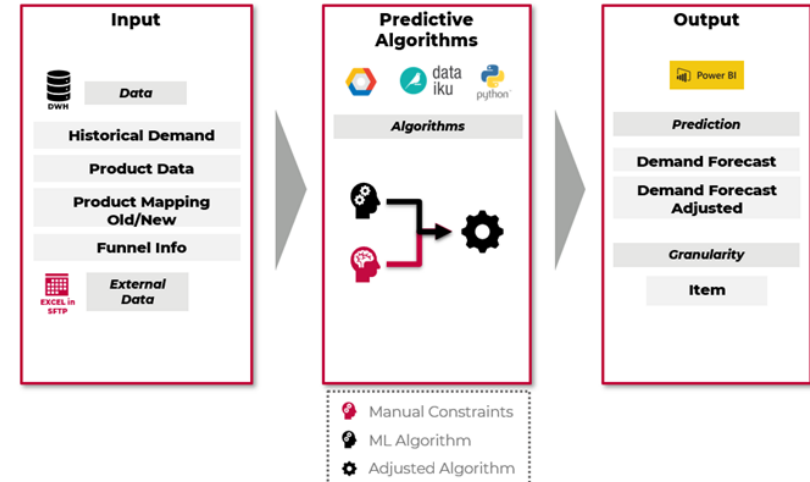


Figure 2. End-to-end workflow: input data, XGBoost algorithm and Power BI output.

5. Assumed Input Features

Demand history	Product context	Business signals	Forecast output
SKU x market x month, historical demand, seasonality index	category, lifecycle status, old/new product mapping	funnel index, promotion flag, stock/availability flag, S&OP notes	ML forecast, adjusted forecast, MAPE band, exception flag

6. Model Design and Hybrid Forecasting Logic

XGBoost Forecasting Layer

- Selected because it handles structured, nonlinear business data and can combine historical demand with product, lifecycle and market features.
- Assumed validation used rolling backtests and accuracy metrics such as MAPE, MAE and RMSE.
- Business guardrails were applied for product transitions, availability constraints and exceptional demand signals.

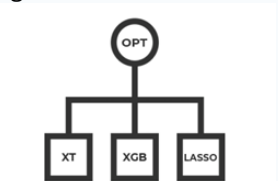


Figure 3. Simplified optimizer/ensemble concept from draft.

Hybrid Process Logic

Layer	Description	Owner
ML forecast	XGBoost forecast generated for SKU-level demand	IT + research support
Excel baseline	Traditional statistical forecast as comparison layer	SCM + project owner
Forecast combination	ML and baseline compared/combined for stability	Project owner + SCM
Planner adjustment	Pareto-based review of key SKUs and exceptions	Planners / SCM

7. Excel Planning Layer

Excel was retained as a familiar planning interface for structured review, scenario checks and manual exception handling. Sensitive values and identifiers are blurred.

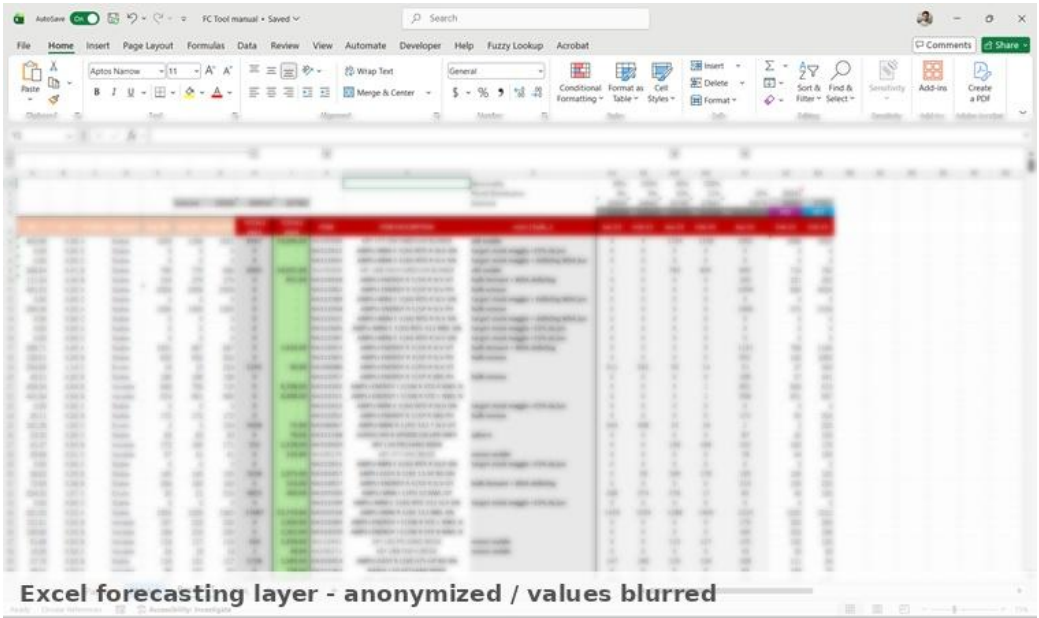


Figure 4. Anonymized Excel forecasting layer from uploaded project draft.
Anonymized/synthetic content. No source code, production data, credentials or proprietary logic included.

8. Planner Adjustment, Power BI Output and Impact

The tool used Pareto segmentation to direct planner attention to the most important exceptions. Around 20% of SKUs drove roughly 80% of business impact, allowing planners to focus on approximately 100 key items instead of manually reviewing 1,000+ SKUs.

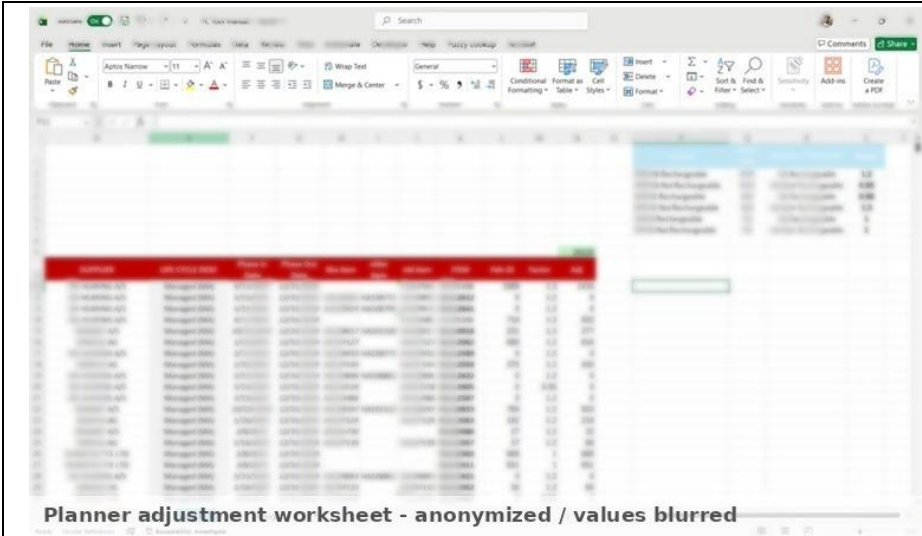


Figure 5. Planner adjustment worksheet, anonymized and blurred.

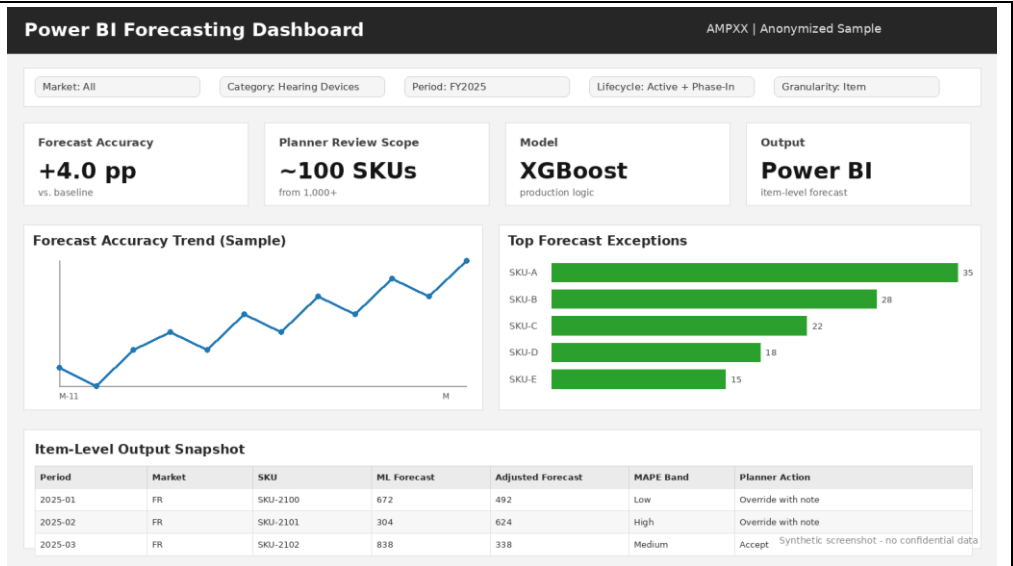


Figure 6. Synthetic Power BI-style output dashboard.

Metric / Outcome	Result	Business Meaning
Forecast accuracy	Improved by approx. 4%	Better reliability at volume level
Planner workload	1,000+ SKUs to ~100 key items	More exception-based decision-making
Segmentation insight	20% of SKUs drove ~80% impact	Supports Pareto review process
Process scalability	Hybrid Excel + XGBoost + Power BI	More repeatable forecasting workflow

Confidentiality statement: All screenshots, sample data and technical descriptions are anonymized or synthesized for portfolio use.